AI Agent Research Compendium — Volume 2: Memory & RAG for Production Agents

> The most impactful papers on agent memory architectures and retrieval-augmented generation, filtered for practical applicability.

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1. Agent Memory Architectures

Rethinking Memory Mechanisms of Foundation Agents (Survey)

- **arXiv:** 2602.06052 | **PDF:** https://arxiv.org/pdf/2602.06052v3
- **Key Insight:** Comprehensive survey of agent memory by substrate (internal/external), cognitive mechanism (episodic, semantic, working, procedural), and subject. **Essential reading.**

Continuum Memory Architectures for Long-Horizon LLM Agents

- **arXiv:** 2601.09913 | **PDF:** https://arxiv.org/pdf/2601.09913v1
- **Key Insight:** Persistent, temporally chained internal state instead of stateless RAG lookups. **Must-read for production agent design.**

MAGMA: Multi-Graph based Agentic Memory Architecture

- **arXiv:** 2601.03236 | **PDF:** https://arxiv.org/pdf/2601.03236v1
- **Key Insight:** Multi-graph memory across semantic, temporal, causal, and entity graphs with policy-guided traversal. **Production-relevant.**

E-mem: Multi-agent Episodic Context Reconstruction

- **arXiv:** 2601.21714 | **PDF:** https://arxiv.org/pdf/2601.21714v1
- **Key Insight:** Episodic memory framework where assistant agents maintain uncompressed contexts while master agent orchestrates global planning.

AMA: Adaptive Memory via Multi-Agent Collaboration

- **arXiv:** 2601.20352 | **PDF:** https://arxiv.org/pdf/2601.20352v2
- **Key Insight:** Multi-agent memory with hierarchical granularity, adaptive query routing, consistency verification, and targeted refresh.

AtomMem: Learnable Dynamic Agentic Memory with Atomic Memory Operation

- **arXiv:** 2601.08323 | **PDF:** https://arxiv.org/pdf/2601.08323v2
- **Key Insight:** Decomposes memory management into atomic CRUD operations, learns autonomous policy via SFT + RL.

ProcMEM: Learning Reusable Procedural Memory via Non-Parametric PPO

- **arXiv:** 2602.01869 | **PDF:** https://arxiv.org/pdf/2602.01869v1
- **Key Insight:** Agents save procedural skills from past runs and reuse them — reduces repeated computation in production.

FadeMem: Biologically-Inspired Forgetting for Efficient Agent Memory

- **arXiv:** 2601.18642 | **PDF:** https://arxiv.org/pdf/2601.18642v2
- **Key Insight:** Adaptive exponential decay, LLM-guided conflict resolution, and dual-layer hierarchy.

SwiftMem: Fast Agentic Memory via Query-aware Indexing

- **arXiv:** 2601.08160 | **PDF:** https://arxiv.org/pdf/2601.08160v1

- **Key Insight:** Sub-linear retrieval via DAG-Tag indexing with embedding-tag co-consolidation.

SimpleMem: Efficient Lifelong Memory for LLM Agents

- **arXiv:** 2601.02553 | **PDF:** https://arxiv.org/pdf/2601.02553v3
- **Key Insight:** Three-stage: semantic lossless compression, online semantic synthesis, intent-aware retrieval planning.

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2. Advanced RAG Systems

Corpus2Skill: Don't Retrieve, Navigate

- **arXiv:** 2604.14572 | **PDF:** https://arxiv.org/pdf/2604.14572
- **Key Insight:** Compiles corpus offline into hierarchical skill tree, replacing retrieval with skill-tree traversal. **Novel enterprise approach.**

JADE: Bridging Strategic-Operational Gap in Dynamic Agentic RAG

- **arXiv:** 2601.21916 | **PDF:** https://arxiv.org/pdf/2601.21916v1
- **Key Insight:** Joint planning-execution optimization as cooperative multi-agent team with shared backbone and outcome-based rewards.

ProRAG: Process-Supervised RL for RAG

- **arXiv:** 2601.21912 | **PDF:** https://arxiv.org/pdf/2601.21912v1
- **Key Insight:** MCTS-based step-level rewards identify and fix flawed reasoning steps in multi-hop retrieval.

To Retrieve or To Think? Agentic Context Evolution

- **arXiv:** 2601.08747 | **PDF:** https://arxiv.org/pdf/2601.08747v2
- **Key Insight:** Dynamically decides whether to retrieve or reason at each step, eliminating redundant retrieval.

L-RAG: Entropy-Based Lazy Loading for RAG

- **arXiv:** 2601.06551 | **PDF:** https://arxiv.org/pdf/2601.06551v1

- **Key Insight:** Entropy-based gating bypasses vector DB when uncertainty is low, only retrieves when genuinely uncertain.

Less is More for RAG: Information Gain Pruning

- **arXiv:** 2601.17532 | **PDF:** https://arxiv.org/pdf/2601.17532v1
- **Key Insight:** Generator-aligned reranking selects evidence by utility and filters harmful passages before truncation.

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3. Graph-Based Memory & RAG

Deep GraphRAG: Hierarchical Retrieval and Adaptive Integration

- **arXiv:** 2601.11144 | **PDF:** https://arxiv.org/pdf/2601.11144v3
- **Key Insight:** Hierarchical global-to-local retrieval with beam search re-ranking and RL-trained dynamic weighting.

Reliable Graph-RAG for Codebases: AST vs LLM-Extracted KGs

- **arXiv:** 2601.08773 | **PDF:** https://arxiv.org/pdf/2601.08773v1
- **Key Insight:** Benchmarks vector-only, LLM-extracted KG, and AST-derived graph pipelines for code RAG. **Essential for code agents.**

Relink: Query-Driven Evidence Graph On-the-Fly

- **arXiv:** 2601.07192 | **PDF:** https://arxiv.org/pdf/2601.07192v1
- **Key Insight:** Dynamically builds query-specific evidence graphs from latent relation pool, discarding distractors.

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4. Memory Optimization & Efficiency

BudgetMem: Query-Aware Budget-Tier Routing

- **arXiv:** 2602.06025 | **PDF:** https://arxiv.org/pdf/2602.06025v1
- **Key Insight:** Routes memory queries to different processing tiers based on difficulty — controls cost-accuracy trade-off.

ShardMemo: Masked MoE Routing for Sharded Memory

- **arXiv:** 2601.21545 | **PDF:** https://arxiv.org/pdf/2601.21545v1
- **Key Insight:** Tiered memory service using masked MoE routing to probe only eligible shards under budget.

Active Context Compression: Autonomous Memory Management

- **arXiv:** 2601.07190 | **PDF:** https://arxiv.org/pdf/2601.07190v1
- **Key Insight:** Physarum-inspired architecture where agent autonomously consolidates and prunes interaction history.

Learning How to Remember: Meta-Cognitive Memory Management

- **arXiv:** 2601.07470 | **PDF:** https://arxiv.org/pdf/2601.07470v1
- **Key Insight:** Treats memory abstraction as learnable skill, training memory copilot via DPO for structuring and reusing memories.

This volume covers 20 papers from Memory & RAG. Selected for novel architectures, production-relevant optimizations, and graph-based approaches scaling beyond simple vector retrieval.